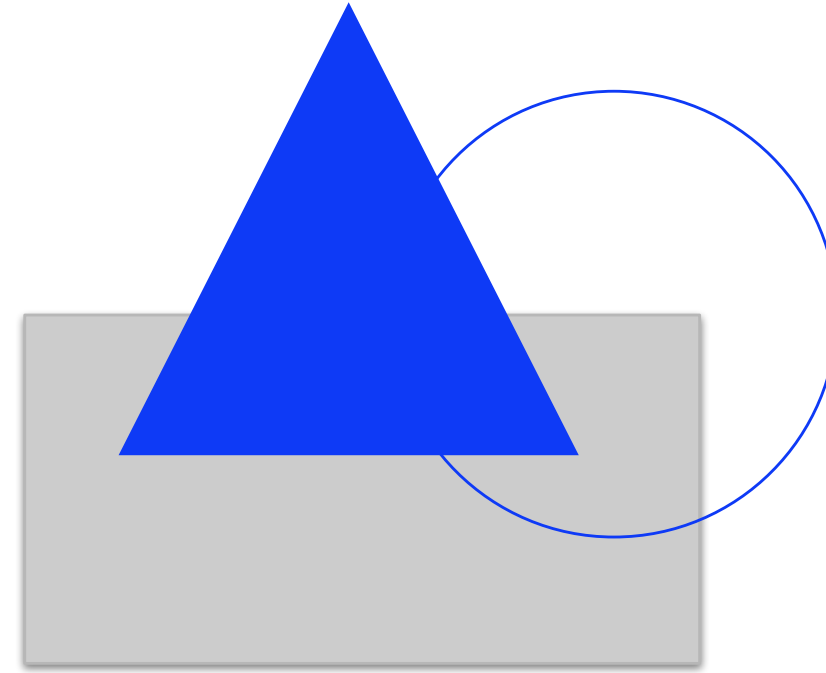




Chinese Abstract Landscape Painting GAN Colorization Task

Tiep Tran, Chau Nguyen, Deepak Anthony David, Viet Ton, Anju Varghese

- Reference**
- [1]CycleGAN 2017, [arxiv:1703.10593](#)
 - [2]DLP-GAN 2024, [arXiv:2403.03456](#)
 - [3]Xue 2020, [arXiv:2011.05552](#) [[cs.CV](#)]
 - [4]Rougetet 2020, [kaggledataset](#)

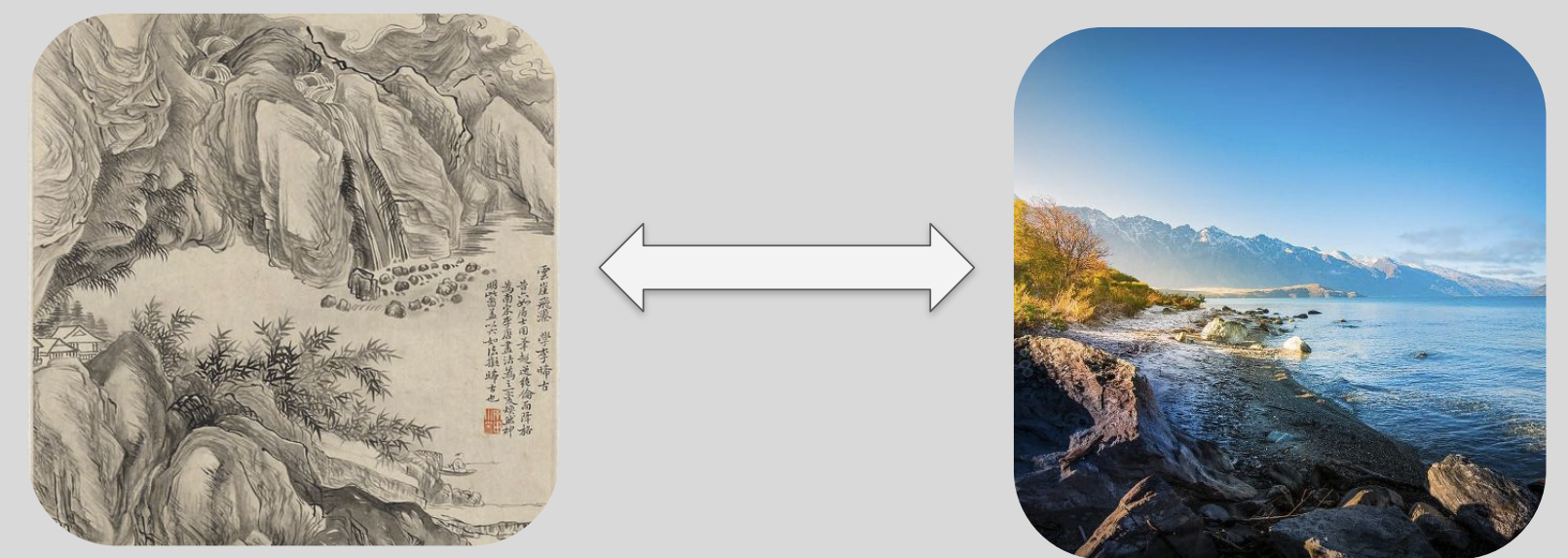
01. Background



- Previous style transfer researches focus on translating photograph into western-styled oil painting (mimics Van Gogh, Monet, ...) while **Chinese landscape art (Shan Shui)** is a very unique style of painting, involving abstract representation of landscape via expressive, free use of ink and brush strokes.
- In this project, we aim to address the **colorization task of Chinese style landscape painting using GAN**. Specifically we will be experimenting with modified **CycleGAN**[1] and **Draw Modern Chinese Landscape Photos GAN (DLP-GAN)**[2].

02. Challenge

- The primary difficulty lies in the Information Asymmetry between the source and target domains:
- **The Domain Gap**: We are translating from an Information-Poor domain (abstract, monochromatic sketches) to an Information-Rich domain (detailed, colorful photos).
 - **Texture Hallucination**: The model cannot simply "color in" the lines; it must hallucinate missing high-frequency details—such as rock textures, water ripples, and vegetation—that do not exist in the sparse input.
 - **Unpaired Data**: No ground-truth photographs exist for ancient paintings, requiring unsupervised learning techniques



03. Project Aims

- This project aims to bridge the gap between abstract ink art and photorealism by:
- **Implementing DLP-GAN**: A specialized GAN framework designed to hallucinate modern landscape photos from ancient Chinese paintings.
 - **Architecture Comparison**: Benchmarking this specialized architecture against a CycleGAN baseline.
 - **Optimization**: Modifying loss weights to prioritize texture synthesis and realistic colorization over strict content constraint, moving beyond simple "color tinting."

07. Future Works

- Current models need more experiment with hyperparameters and better handling of artifact (letter, seal)
- Transfer learning - How well would these models perform on other domains (Manga, Infrared images,...)
- Experiment with Diffusion models
- Experiment with painting restoration task

04. Training Configs



- Using 2192 high quality Chinese landscape paintings dataset, Xue2020[3] as Domain A
- 2000 high quality landscape photo collected from Kaggle[4] as Domain B

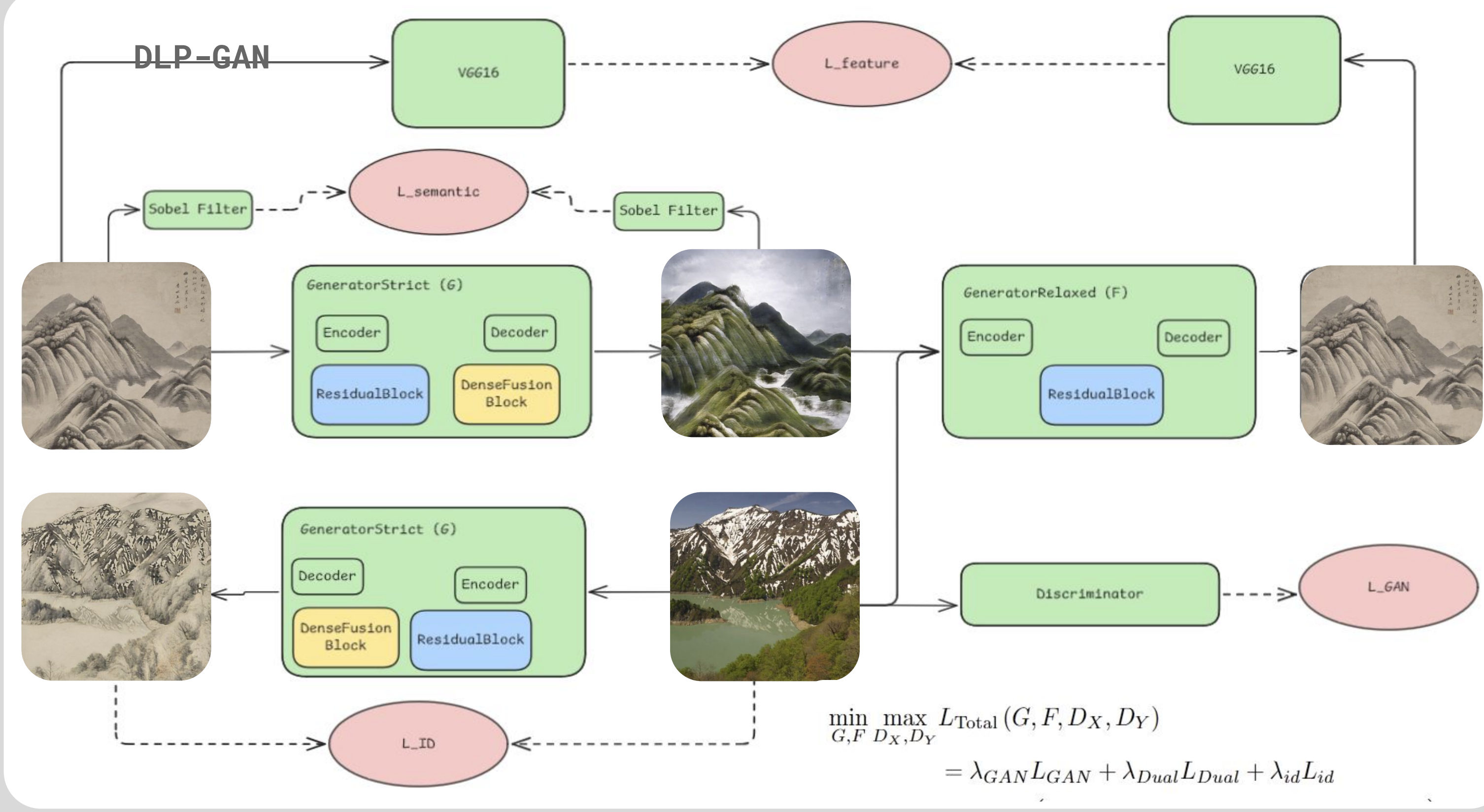
CycleGAN
Train on a subset of data (600 samples) for 5-6 hours

- EPOCHS = 200
- LR = 0.0002 (Linearly decayed to 0 for the last 100 epochs)
- BATCH_SIZE = 1
- INPUT_DIM = 512x512
- $\lambda_{GAN}=1$, $\lambda_{CYCLE}=10$, $\lambda_{ID}=5$
- Image Pool (50 images) for stabilization

DLP-GAN
Train on 1 RTX Ada6000 for 20 hours

- EPOCHS = 200
- LR = 0.0002 (Adam, $\beta_1=0.5$, $\beta_2=0.999$)
- BATCH_SIZE = 2 (Instance Norm)
- INPUT_DIM = 512x512
- $\lambda_{GAN} = 4$ (1 in original paper)
- $\lambda_{DUAL} = 8$ (10 in original paper)
- $\lambda_{ID} = 3$ (5 in original paper)

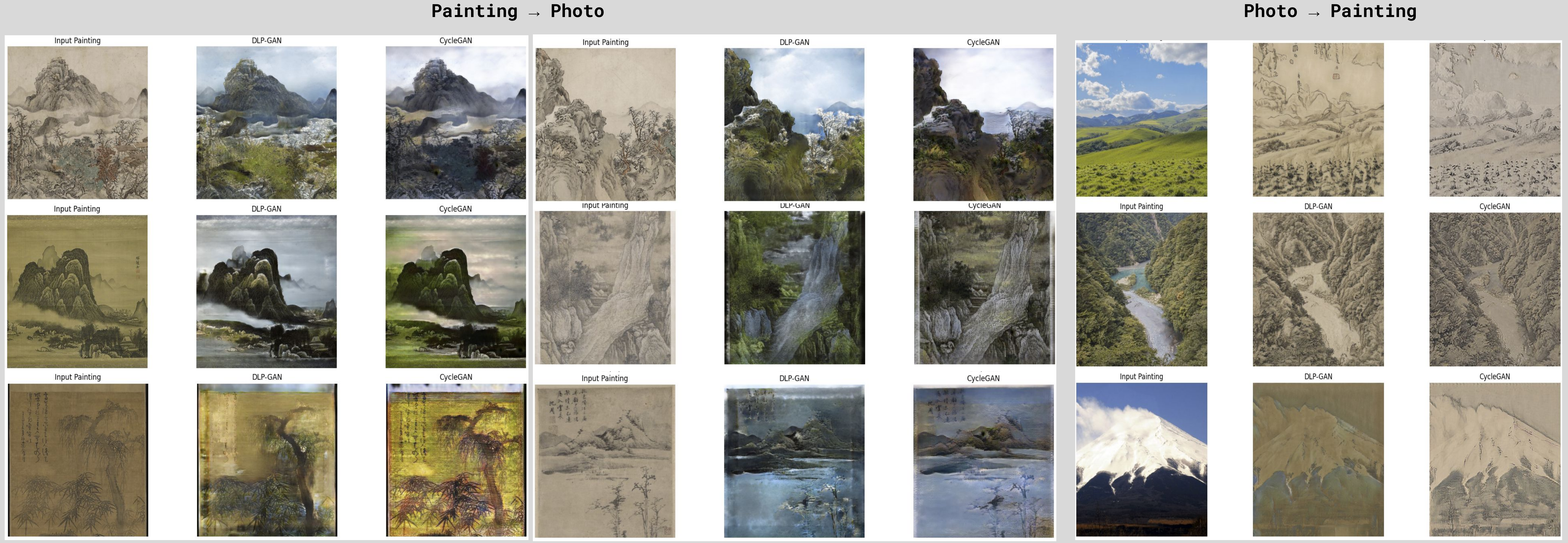
05. Methods



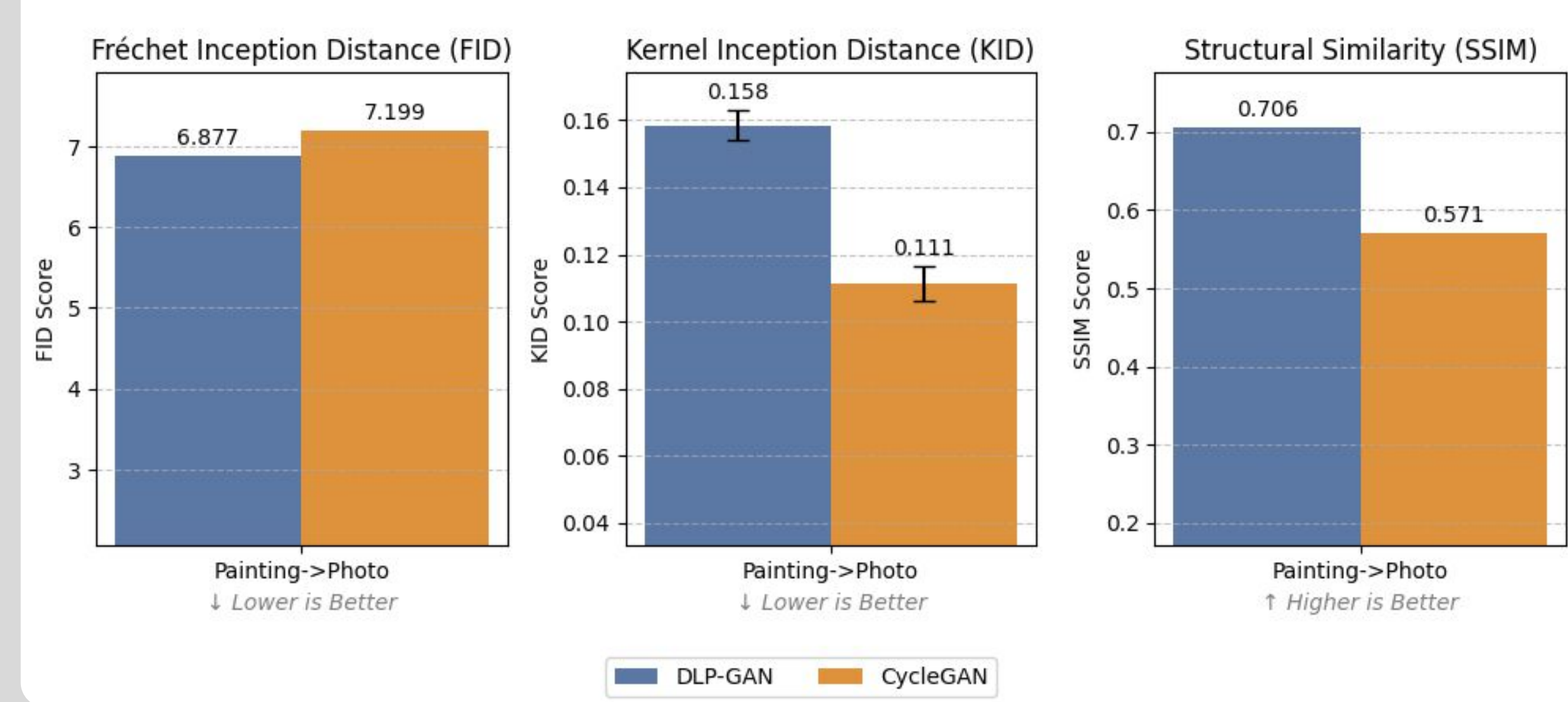
We adopted an Asymmetric Cycle-Consistent framework:

- Generator Architecture:
 - **Strict Generator (G:Painting-Photo)**: Incorporated a Dense Fusion Block to maximize feature reuse. This allows the network to propagate semantic edge information deep into the network to guide texture generation.
 - **Relaxed Generator (F:Photo-Painting)**: Utilized a ResNet-based generator with 9 Residual Blocks to handle 512px resolution.
- Re-Balanced Objectives:
 - **Adversarial Loss ($\lambda_{GAN}=4$)**: Significantly boosted (vs. paper default of 1) to force the generator to prioritize realism and fool the discriminator.
 - **Identity Loss ($\lambda_{ID}=3$)**: Reduced (vs. paper default of 5) to allow the model to break away from the monochromatic input histogram and generate vibrant colors.
 - **Dual-Consistency Loss**: Enforced consistency in VGG features and Semantic edges rather than pixels.

06. Preliminary Results



Model Evaluation Metrics Comparison: DLP-GAN vs CycleGAN



Quantitative Success:

- SSIM (0.706): Our DLP-GAN outperformed CycleGAN (0.571), indicating better preservation of structural content (mountains/trees).
- FID (6.877): We achieved a lower distance score than CycleGAN (7.199), indicating our generated distribution is statistically closer to real photos.

Qualitative Observations:

- Texture Synthesis: DLP-GAN successfully generated mossy rock textures and water reflections where the baseline produced noise.
- Artifact Removal: In some samples, DLP-GAN was able to remove letters and red seals.
- Atmospheric Depth: The model correctly interpreted ink washes as volumetric mist/clouds.